

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME : Cryptography and Network Security

COURSE CODE : CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Lekkala Dattu Kumar/192324141 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student : L Dattu Kumar

Annexure A

APPLICATION- BASED QUESTIONS

3. Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

Given:

- Plaintext: HELLO

- Keyword: KEY
- Technique: **Vigenère Cipher (keyword-based encryption)**

a. Encryption process step-by-step

Using the standard mapping:

A = 0, B = 1, C = 2, ..., Z = 25

Repeat the keyword until it matches the length of the message:

Plaintext	H	E	L	L	O
Value	7	4	11	11	14
Keyword	K	E	Y	K	E
Key Value	10	4	24	10	4
Calculation	$(7+10) \bmod 26$	$(4+4) \bmod 26$	$(11+24) \bmod 26$	$(11+10) \bmod 26$	$(14+4) \bmod 26$
Ciphertext	R	I	J	V	S

Therefore:

Plaintext: HELLO

Repeated Key: KEYKE

Ciphertext: RIJVS

Step-by-step calculation

1. H + K

- $H = 7, K = 10$
- $(7 + 10) \bmod 26 = 17$
- $17 = \mathbf{R}$

2. E + E

- $E = 4, E = 4$
- $(4 + 4) \bmod 26 = 8$
- $8 = \mathbf{I}$

3. **L + Y**

- $L = 11, Y = 24$
- $(11 + 24) \bmod 26 = 35 \bmod 26 = 9$
- $9 = \mathbf{J}$

4. **L + K**

- $L = 11, K = 10$
- $(11 + 10) \bmod 26 = 21$
- $21 = \mathbf{V}$

5. **O + E**

- $O = 14, E = 4$
- $(14 + 4) \bmod 26 = 18$
- $18 = \mathbf{S}$

Final Result

HELLO $\rightarrow$ RIJVS

b. Influence of the repeating keyword

The keyword KEY is repeated to match the length of the plaintext:

HELLO

KEYKE

Each plaintext character is combined with the corresponding keyword character.

- H is encrypted using K $\rightarrow$ R
- E is encrypted using E $\rightarrow$ I
- L is encrypted using Y $\rightarrow$ J
- L is encrypted using K $\rightarrow$ V
- O is encrypted using E $\rightarrow$ S

Thus, the repeating keyword determines the **shift applied to each plaintext character**.

Because different key characters produce different shifts, identical plaintext characters can produce different ciphertext characters.

For example, the two L characters in HELLO become **J** and **V** because they are encrypted using different key characters (Y and K).

5. Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Given

ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

This is a **transposition cipher** where the letters are rearranged rather than substituted.

The most probable original message is:

WE ARE DISCOVERED. FLEE AT ONCE

Removing spaces and punctuation:

WEAREDISCOVEREDFLEEATONCE

a. Reversing the encryption process

The received ciphertext is:

WECRLTEERDSOEFEAOCAIVDEN

The transposition process has rearranged the letters of the original message.

By reconstructing the character arrangement and reading the characters in their original order, we obtain:

Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

Decrypted message:

WEAREDISCOVEREDFLEEATONCE

Adding spaces:

"WE ARE DISCOVERED. FLEE AT ONCE."

Decryption result

Ciphertext

Plaintext

WECRLTEERDSOEFEAOCAIVDEN WE ARE DISCOVERED. FLEE AT ONCE

The important point is that a **transposition cipher does not change the actual letters**. It only changes their positions.

b. Why this method is vulnerable in modern communication

A basic transposition cipher is not sufficiently secure for modern communication because:

- 1. It preserves the original letters.**

The characters are only rearranged, making statistical analysis easier.

- 2. It has a limited key space.**

If the arrangement/key is small, an attacker can try different permutations until meaningful plaintext is obtained.

- 3. It is vulnerable to brute-force attacks.**

Modern computers can test a large number of possible arrangements quickly.

- 4. It provides weak confidentiality.**

The same plaintext encrypted using the same arrangement can produce the same ciphertext.

- 5. It does not provide modern security properties.**

A simple transposition cipher does not provide strong authentication or integrity .

Therefore, although transposition methods were useful historically, they are **not suitable as a standalone encryption mechanism for modern corporate communication**.

c. Practical enhancement

A practical enhancement is to **replace the basic transposition cipher with a modern symmetric encryption algorithm such as AES.**

For example:

Original system:

Plaintext → Transposition Cipher → Ciphertext

Improved system:

Plaintext → AES Encryption + Secure Key → Ciphertext

AES provides substantially stronger confidentiality than a simple transposition technique. In a real corporate system, an authenticated encryption mode such as **AES-GCM** can additionally provide integrity/authentication of the encrypted data.

Final Conclusion

The legacy transposition method can be reversed because it only rearranges characters. For modern corporate communication, it should be replaced with a modern authenticated encryption scheme such as **AES-GCM**, together with secure key management.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts

Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementat ion with minor issues	Basic implementati on with errors or inefficiencies	Incorrect or incomplete implementati on
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct executio n	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	interpretatio n Basic analysis with limited	insights Poor or incorrect ana lysis of results
Documentat ion	10	Clear, well structured documentation with diagrams, code, and results	Organized documentati on with necessary details	Basic documentatio n with missing details	Poor or incomplete documentatio n